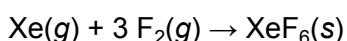


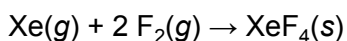
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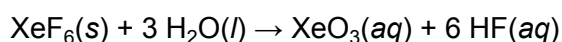


Reaction 1

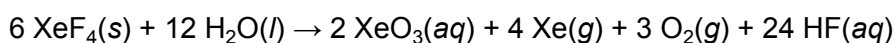


Reaction 2

Dissolving XeF₆(s) or XeF₄(s) in water forms a weakly acidic solution containing XeO₃(aq), as indicated by Reactions 3 and 4.

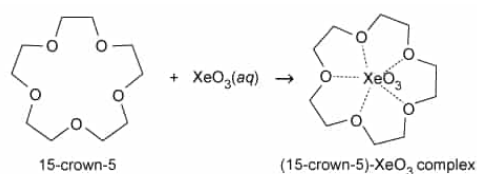


Reaction 3



Reaction 4

XeO₃ is stable in aqueous solution but as a dried solid, XeO₃ is violently explosive and will detonate spontaneously upon heating or mechanical shock. However, adding 15-crown-5 prior to isolation results in the formation of a complex (Reaction 5) that stabilizes XeO₃ and makes the complex safe to isolate and dry.



Reaction 5

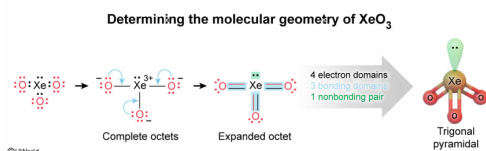
What is the molecular geometry of a XeO_3 molecule?

- ☐ A. Trigonal planar [25%]
☐ B. T-shaped [11%]
✓ ☒ C. Trigonal pyramidal [52%]
☐ D. Tetrahedral [10%]

Correct

51% answered correctly

Explanation:



According to valence-shell electron-pair repulsion (VSEPR) theory, the electron pairs around a central atom in a molecule will adopt an **electron geometry** (ie, a spatial arrangement) that maximizes the separation between the electron domains and minimizes the repulsion between the electrons. Using VSEPR theory, the **molecular geometry** (ie, the shape) of a molecule can be predicted from its Lewis structure by evaluating the orientations that the *bonded atoms* will adopt due to the number of bonding and nonbonding electron domains present around the central atom.

In the formation of XeO_3 , Xe contributes 8 valence electrons, and each of the three O atoms contributes 6 valence electrons (ie, a total of $8 + 6 + 6 + 6 = 26$ valence electrons). This results in a molecular structure with a central Xe atom surrounded by **4 electron domains** (ie, **3 bonding domains** and **1 nonbonding electron pair**). Although the Xe atom can be drawn with an "**expanded octet**" by forming double bonds to minimize charges, the arrangement of electron domains remains the same.

Therefore, VSEPR theory predicts a **tetrahedral electron geometry** but because only three of the electron domains around the Xe atom are bonding domains, the **molecular geometry** is predicted to be **trigonal pyramidal**.

(Choice A) A **trigonal planar** molecular geometry would occur if the Xe atom had 3 electron domains consisting of 3 bonding domains and 0 nonbonding electron pairs. The presence of a lone electron pair in XeO_3 exerts a repulsive force that prevents the Xe–O bonds from being planar.

(Choice B) A **T-shaped** molecular geometry would occur if the Xe atom had 5 electron domains consisting of 3 bonding domains and 2 nonbonding electron pairs.

(Choice D) A **tetrahedral** molecular geometry would result if all 4 electron domains around the Xe atom were involved in bonding (ie, 4 bonding domains and 0 nonbonding electron pairs).

Educational objective:

Valence-shell electron-pair repulsion (VSEPR) theory holds that the electron domains around a central atom will adopt an electron geometry that maximizes the separation and minimizes the repulsion between the electrons. The molecular geometry (shape) of a molecule depends on the electron geometry and number of nonbonding electron pairs of the central atom.

Time spent: 0 sec

Subject:
General Chemistry

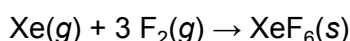
QID: 403394

System:
Interactions of Chemical Substances

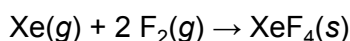
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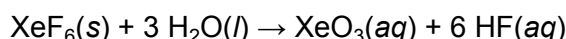


Reaction 1

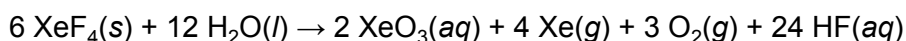


Reaction 2

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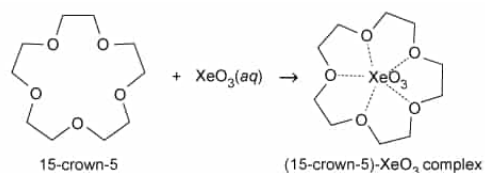


Reaction 3



Reaction 4

XeO_3 is stable in aqueous solution but as a dried solid, XeO_3 is violently explosive and will detonate spontaneously upon heating or mechanical shock. However, adding 15-crown-5 prior to isolation results in the formation of a complex (Reaction 5) that stabilizes XeO_3 and makes the complex safe to isolate and dry.



Reaction 5

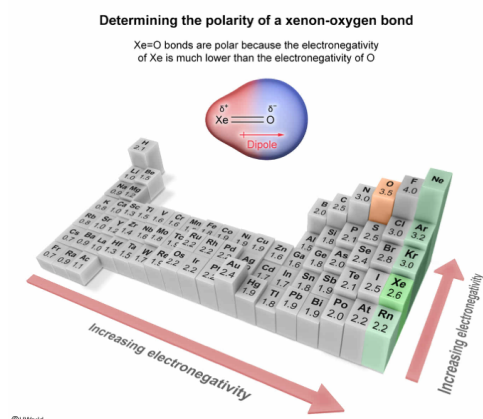
In XeO_3 , the xenon-oxygen bonds are:

- ☐ A. nonpolar, because the electron affinity of Xe is close to zero. [5%]
- ☐ B. nonpolar, because the electronegativities of Xe and O are nearly equal. [6%]
- ☐ C. polar, because the electronegativity of Xe is much higher than the electronegativity of O. [8%]
- ✓ ☒ D. polar, because the electronegativity of Xe is much lower than the electronegativity of O. [79%]

Correct

79% answered correctly

Explanation:



Covalent bonds form when two atoms (usually nonmetals) share electrons. If the **electronegativities** of two covalently bonded atoms are **significantly dissimilar**, then the electrons are **shared unequally** between the bonded atoms and a **dipole** is formed, producing a polar covalent bond. Conversely, if the electronegativities of the bonded atoms are very similar, the bond is nonpolar.

On the periodic table, electronegativity tends to **increase across a row** and **decrease down a column**. Although noble gases are usually excluded from electronegativity tables (because these elements rarely form bonds), the electronegativity trend still holds true for the noble gas group. As such, Xe has a significantly lower electronegativity than O because Xe is a little farther to the right but much lower on the periodic table than O.

Therefore, the Xe=O bonds in XeO_3 are polar because the electronegativity of Xe is much lower than the electronegativity of O.

(Choice A) Although the **electron affinity** of Xe is **close to zero**, this property does not determine the polarity of a chemical bond between two atoms.

(Choice B) The electronegativities of Xe and O cannot be nearly equal because their positions on the periodic table (relative to fluorine) are very different.

(Choice C) Although the Xe=O bonds are polar, the polarity results from the electronegativity of Xe being much *lower than* the electronegativity of O.

Educational objective:

The difference in electronegativity between two bonded atoms determines the polarity of the bond between the atoms. Greater separation between two elements on the periodic table usually indicates a greater difference in electronegativity and a larger dipole moment (higher polarity).

Time spent: 0 sec

Subject:
General Chemistry

QID: 403395

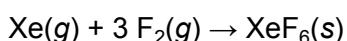
System:
Interactions of Chemical Substances

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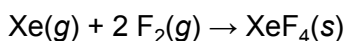
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Under appropriate conditions, it is possible for species that have a very high electron affinity (eg, fluorine, oxygen) to overcome the high ionization energy of noble gas atoms and form a chemical bond. In such cases, the reactivity of the noble gases tends to increase going down the periodic column due to the valence electrons occupying higher-energy orbitals farther from the nucleus. As a result, xenon can form several compounds.

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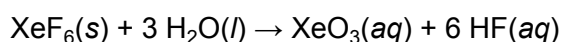


Reaction 1

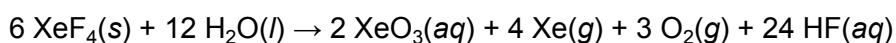


Reaction 2

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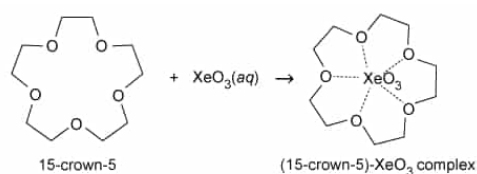


Reaction 3



Reaction 4

XeO_3 is stable in aqueous solution but as a dried solid, XeO_3 is violently explosive and will detonate spontaneously upon heating or mechanical shock. However, adding 15-crown-5 prior to isolation results in the formation of a complex (Reaction 5) that stabilizes XeO_3 and makes the complex safe to isolate and dry.



Reaction 5

Based on its molecular structure, is XeO_3 a polar molecule?

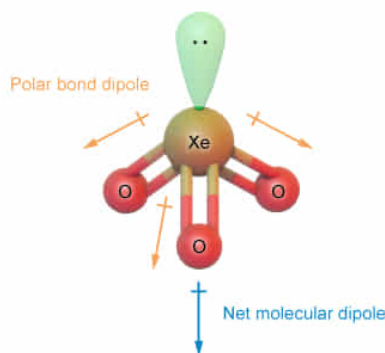
- ☐ A. Yes, because the $\text{Xe}=\text{O}$ bonds are polar and any molecule with polar bonds is a polar molecule. [1%]
- ☐ B. No, because the $\text{Xe}=\text{O}$ bonds are nonpolar. [2%]
- ✓ ☒ C. Yes, because the $\text{Xe}=\text{O}$ bonds are polar and have a geometry that gives the molecule a net dipole. [78%]
- ☐ D. No, because the $\text{Xe}=\text{O}$ bonds are polar but have a geometry that causes the dipoles to cancel. [17%]

Correct

77% answered correctly

Explanation:

Determining the molecular polarity of XeO_3



XeO_3 is a polar molecule because the $\text{Xe}=\text{O}$ bonds are polar and have a geometry that results in a net dipole

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A molecule can be considered polar only if it has a significant net dipole. As such, the polarity of a molecule depends on the number of **polar bonds** present, the **strength of the dipoles** in each polar bond, and the **shape (geometry)** of the molecule. The shape of a molecule affects its overall polarity because its geometry determines the **orientations of the dipoles**, and dipoles that are oriented in opposing directions can partially or fully cancel each other.

According to **valence-shell electron-pair repulsion (VSEPR) theory**, the shape of a molecule can be determined from its **Lewis structure** by distributing the electron domains as far apart as possible around a

central atom to minimize electron repulsion. A molecule's **electron geometry** is determined by the number of electron-dense regions, but the **molecular geometry** (shape) is determined by the orientations the **bonded atoms** adopt due to the number of bonding and **nonbonding electron** domains around the central atom.

The molecular structure of XeO_3 has a central Xe atom surrounded by 4 electron domains (ie, 3 bonding domains and 1 nonbonding electron pair). Accordingly, VSEPR theory predicts the molecular geometry to be **trigonal pyramidal**. The **$\text{Xe}=\text{O}$ bonds are polar** due to a significant difference in electronegativity between the atoms and because the bonds have a partly **complimentary orientation** within the trigonal pyramidal geometry, the bond dipoles partially add together, yielding a **net molecular dipole**.

(Choice A) A molecule with polar bonds may still be nonpolar overall if its shape orients the polar bond dipoles in an arrangement that causes them to completely cancel each other (eg, **CF_4 is nonpolar** but has four polar C–F bonds).

(Choice B) The $\text{Xe}=\text{O}$ bonds are *polar* because the electronegativity of Xe is much lower than that of O.

(Choice D) The three Xe=O bond dipoles would cancel only if the XeO₃ molecule had a trigonal planar geometry.

Educational objective:

Polar molecules have a significant net dipole that depends on the number of polar bonds, the strength of the dipoles, and the shape (geometry) of the molecule. The molecular geometry can be determined from Lewis structures using valence-shell electron-pair repulsion (VSEPR) theory.

Time spent:0 sec

Subject:
General Chemistry

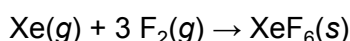
QID:403396

System:
Interactions of Chemical Substances

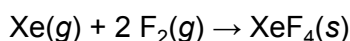
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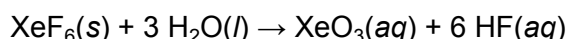


Reaction 1

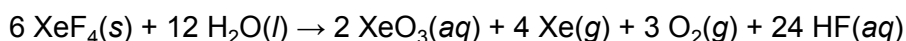


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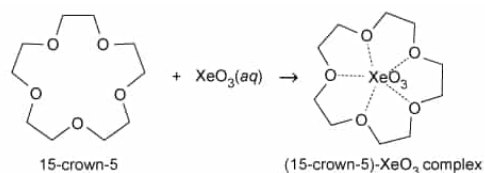


Reaction 3



Reaction 4

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Reaction 5

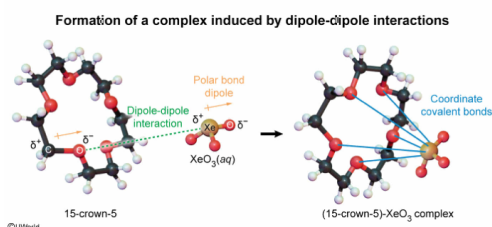
Which of the following intermolecular forces primarily cause the attractions between 15-crown-5 and XeO_3 that result in the formation of the coordinate covalent bonds present in the (15-crown-5)- XeO_3 complex?

- ☐ A. London dispersion forces [5%]
- ✓ ☒ B. Dipole-dipole forces [59%]
- ☐ C. Dipole-induced dipole forces [27%]
- ☐ D. Ion-dipole forces [7%]

Correct

60% answered correctly

Explanation:



A **dipole** consists of opposite electric charges separated across a distance (eg, a chemical bond). In molecules, **permanent dipoles** form across some **covalent bonds** due to **unequal sharing of electrons** between two atoms caused by significant differences in **electronegativity**. In a dipole, the region with greater electron density (ie, the more electronegative atom) gains a partial negative charge (δ^-), and the region with less electron density (ie, the less electronegative atom) acquires a partial positive charge (δ^+).

Many **intermolecular forces** (ie, noncovalent interactions between atoms and molecules) involve the partial charges of bond dipoles. A **dipole-dipole interaction** occurs when the opposite partial charges of the permanent dipoles of the polar bonds in neighboring molecules form a mutual attraction. In the structures of 15-crown-5 and XeO_3 , the C–O and Xe=O bonds are all polar and have permanent dipoles. As such, the opposite partial charges of these bonds can align and participate in dipole-dipole interactions.

Therefore, primarily dipole-dipole forces cause the mutual attractions between 15-crown-5 and XeO_3 . As the atoms are drawn closer together by these attractions, coordinate covalent bonds are formed and result in the production of the (15-crown-5)- XeO_3 complex.

(Choice A) **London dispersion forces** form when two *nonpolar bonds* induce *momentary dipoles* in each other by distortions in the electron cloud around the atoms.

(Choice C) A **dipole-induced dipole interaction** occurs when the permanent dipole of a polar bond distorts the electron cloud of an atom in a neighboring nonpolar bond and induces a weak temporary dipole. The interactions between 15-crown-5 and XeO_3 involve only polar bonds with *permanent dipoles*.

(Choice D) An **ion-dipole interaction** occurs when the full charge of an ion is attracted to the opposite partial charge of a polar bond dipole. Neither 15-crown-5 nor XeO_3 contain an ionic charge.

Educational objective:

Ion-dipole interactions occur between the partial charge of a permanent dipole and the full charge of an

ion. Dipole–induced dipole interactions occur when a permanent dipole induces a weak temporary dipole in a nonpolar bond or atom. Dipole-dipole interactions occur between the partial charges of two permanent dipoles. London dispersion forces occur between momentary induced dipoles caused by orbital distortions.

Time spent:0 sec

Subject:
General Chemistry

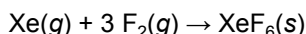
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System:
Interactions of Chemical Substances

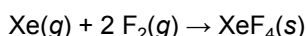
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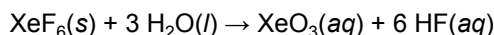


Reaction 1

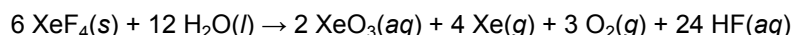


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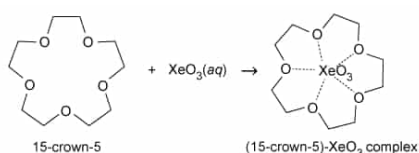


Reaction 3



Reaction 4

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Reaction 5

If 41.4 mg of $\text{XeF}_4(s)$ is hydrolyzed in 5.00 mL of water according to Reaction 4, what is the theoretical yield of $\text{HF}(aq)$ expressed as the molar concentration in the resulting solution? (Note: The molar mass of XeF_4 is 207.28 g/mol.)

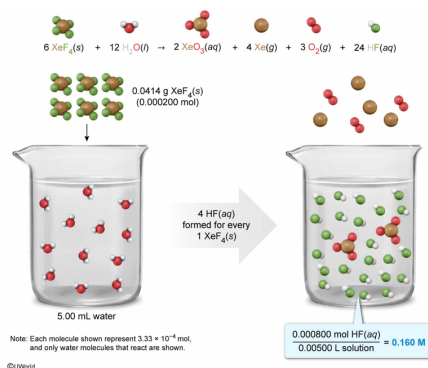
- ☐ A. $8.00 \times 10^{-4} \text{ M}$ [14%]
☐ B. $1.00 \times 10^{-2} \text{ M}$ [10%]
☐ C. $4.00 \times 10^{-2} \text{ M}$ [23%]
☒ D. $1.60 \times 10^{-1} \text{ M}$ [52%]

Correct

51% answered correctly

Explanation:

Theoretical yield of $\text{HF}(aq)$ expressed as a molar concentration



The theoretical yield of a chemical reaction is the maximum amount of a product that can be produced during the reaction from the given amounts of the reactants initially present. The amount of the theoretical yield is determined by the **limiting reactant** (ie, the reactant that is used up first) because once the limiting reactant is gone, the reaction stops and no more products can be formed.

In Reaction 4, water is both a reactant and the solvent. As such, the water is in excess, which makes $\text{XeF}_4(\text{s})$ the limiting reactant. If all a 41.4 mg sample of $\text{XeF}_4(\text{s})$ is converted to product, the number of moles of $\text{XeF}_4(\text{s})$ that react is found by dividing this mass (converted to grams) by the **molar mass** of $\text{XeF}_4(\text{s})$.

$$41.4 \text{ mg } \text{XeF}_4(\text{s}) \times \frac{1 \text{ g } \text{XeF}_4(\text{s})}{1000 \text{ mg } \text{XeF}_4(\text{s})} \times \frac{1 \text{ mol } \text{XeF}_4(\text{s})}{207.28 \text{ g } \text{XeF}_4(\text{s})} = 0.000200 \text{ mol } \text{XeF}_4(\text{s})$$

Reaction 4 shows that 24 moles of $\text{HF}(\text{aq})$ form from every 6 moles of $\text{XeF}_4(\text{s})$ that react (ie, a 4:1 ratio). Applying this **mole ratio** gives the theoretical number of moles of $\text{HF}(\text{aq})$ formed in the resulting solution.

$$0.000200 \text{ mol } \text{XeF}_4(\text{s}) \times \frac{24 \text{ mol } \text{HF}(\text{aq})}{6 \text{ mol } \text{XeF}_4(\text{s})} = 0.000800 \text{ mol } \text{HF}(\text{aq})$$

The **molar concentration** (ie, **molarity**) of a species in solution is defined as the moles of the dissolved species per *liter* of solution. Therefore, dividing the moles of $\text{HF}(\text{aq})$ by the volume of the solution (converted to liters) gives the theoretical yield of $\text{HF}(\text{aq})$ expressed in terms of its molarity.

$$\text{Molar concentration (M)} = \frac{\text{moles of solute}}{\text{solution volume (in liters)}}$$

$$\text{Molarity } \text{HF}(\text{aq}) = \frac{0.000800 \text{ mol } \text{HF}(\text{aq})}{0.00500 \text{ L solution}} = \frac{8.00 \times 10^{-4} \text{ mol}}{5.00 \times 10^{-3} \text{ L}} = \left(\frac{8.00}{5.00}\right) \times 10^{[-4 - (-3)]} \frac{\text{mol}}{\text{L}} \Rightarrow 1.60 \times 10^{-1} \text{ M}$$

(Choice A) A value of 8.00×10^{-4} is the number of *moles* rather than molarity (ie, the moles *per liter*).

(Choice B) A value of 1.00×10^{-2} is obtained if the reaction mole ratio of $\frac{24 \text{ mol } \text{HF}(\text{aq})}{6 \text{ mol } \text{XeF}_4(\text{s})}$ is mistakenly inverted as $\frac{6 \text{ mol } \text{XeF}_4(\text{s})}{24 \text{ mol } \text{HF}(\text{aq})}$ during the second step of the calculation.

(Choice C) A value of $4.00 \times 10^{-2} \text{ M}$ is calculated by incorrectly assuming a 1:1 reaction ratio between $\text{HF}(\text{aq})$ and $\text{XeF}_4(\text{s})$ during the second step of the calculation.

Educational objective:

The theoretical yield is the maximum amount of a product that can be produced from a given amount of the limiting reactant initially present. The number of moles of a product formed is proportional to the number of moles of the reactant species according to the ratios from a balanced reaction. Molar concentrations are defined as the moles of a species per liter of solution.

Time spent: 0 sec

Subject:
General Chemistry

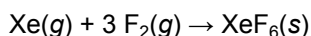
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System:
Interactions of Chemical Substances

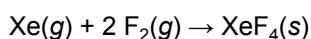
The elements of group 18 on the periodic table are called noble gases because they tend to be unreactive with most elements under most conditions. This lack of reactivity results from the noble gases having a full valence shell, which makes participating in a reaction by gaining, losing, or sharing electrons energetically unfavorable. Accordingly, noble gases have a very high ionization energy and a very low electron affinity.

Under appropriate conditions, it is possible for species that have a very high electron affinity (eg, fluorine, oxygen) to overcome the high ionization energy of noble gas atoms and form a chemical bond. In such cases, the reactivity of the noble gases tends to increase going down the periodic column due to the valence electrons occupying higher-energy orbitals farther from the nucleus. As a result, xenon can form several compounds.

Reactions between xenon and fluorine form relatively stable compounds but the specific compound produced depends on the ratio of these gases. A mixture of Xe(g) and excess $\text{F}_2\text{(g)}$ produces $\text{XeF}_6\text{(s)}$ according to Reaction 1, but a mixture with a 1:2 ratio of Xe(g) to $\text{F}_2\text{(g)}$ results in the formation of $\text{XeF}_4\text{(s)}$ according to Reaction 2.

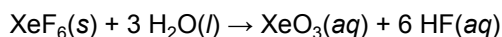


Reaction 1

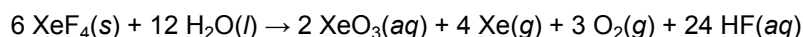


Reaction 2

Dissolving $\text{XeF}_6\text{(s)}$ or $\text{XeF}_4\text{(s)}$ in water forms a weakly acidic solution containing $\text{XeO}_3\text{(aq)}$, as indicated by Reactions 3 and 4.

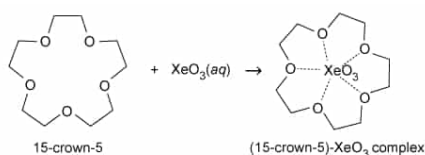


Reaction 3



Reaction 4

XeO_3 is stable in aqueous solution but as a dried solid, XeO_3 is violently explosive and will detonate spontaneously upon heating or mechanical shock. However, adding 15-crown-5 prior to isolation results in the formation of a complex (Reaction 5) that stabilizes XeO_3 and makes the complex safe to isolate and dry.



Reaction 5

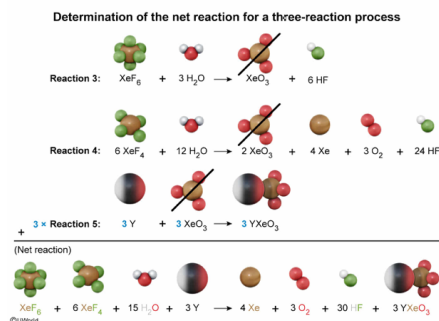
Which of the following is the net reaction for the hydrolysis and complexation of a mixture of 6 moles XeF_4 and 1 mole of XeF_6 following Reactions 3, 4, and 5? (Note: $\text{Y} = 15\text{-crown-5}$.)

- ✓ ☒ A. $\text{XeF}_6 + 6 \text{XeF}_4 + 15 \text{H}_2\text{O} + 3 \text{Y} \rightarrow 4 \text{Xe} + 3 \text{O}_2 + 30 \text{HF} + 3 \text{YXeO}_3$ [59%]
- ☐ B. $\text{XeF}_6 + 6 \text{XeF}_4 + 15 \text{H}_2\text{O} + \text{Y} \rightarrow 6 \text{Xe} + 3 \text{O}_2 + 30 \text{HF} + \text{YXeO}_3$ [22%]
- ☐ C. $6 \text{XeF}_4 + 12 \text{H}_2\text{O} + \text{Y} + \text{XeO}_3 \rightarrow 4 \text{Xe} + 3 \text{O}_2 + \text{YXeO}_3 + 24 \text{HF}$ [7%]
- ☐ D. $\text{XeF}_6 + 6 \text{XeF}_4 + 15 \text{H}_2\text{O} \rightarrow 3 \text{XeO}_3 + 4 \text{Xe} + 3 \text{O}_2 + 30 \text{HF}$ [9%]

Correct

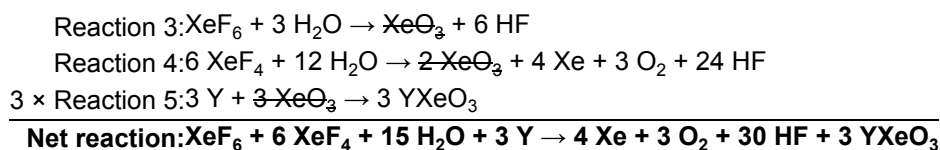
59% answered correctly

Explanation:



A chemical process can sometimes involve more than one reaction step. During the reaction steps, some of the products formed in one step may be consumed as reactants in a later step. As such, the **overall (net) reaction** for the process can be determined from the **sum of the reactions for each step**, and species appearing in equal numbers on both sides of the summation cancel. Because reactions are usually written using the lowest ratio of the chemical species, **multiples** of one (or more) of the reactions may be needed to achieve a **balanced** net reaction.

For a chemical process involving the hydrolysis and complexation of a mixture of 6 moles XeF_4 and 1 mole of XeF_6 , the net reaction for the process is given by the sum of Reactions 3, 4, and 5. However, Reaction 5 must multiplied by 3 to maintain an atom balance because 3 moles of $\text{XeO}_3(aq)$ are produced from Reactions 3 and 4. Therefore, summing the species on each side of the three reactions and canceling species present in equal numbers on both sides yields the correct overall reaction:



(Choice B) This reaction is not balanced (ie, there are 15 O atoms on the reactants side and 9 O atoms on the products side). This result is obtained if Reaction 5 is not multiplied by 3 and counting errors are made with Xe when evaluating the reaction summation.

(Choice C) This is the net reaction resulting from Reactions 4 and 5 only.

(Choice D) This is the net reaction resulting from Reactions 3 and 4 only.

Educational objective:

The balanced net reaction for a chemical process involving more than one step may be found by taking the sum of the reactions for each step. Multiples of one or more reactions may be needed to achieve an overall atom balance. Species matching in type and number on both sides of the summation cancel from the net reaction.

Time spent: 0 sec
Subject:
General Chemistry

QID: 403399
System:
Interactions of Chemical Substances